Operating Systems
Assignment 4 – *Hard*

Instructions:

1. The assignment has to be done solo.
2. You can use Piazza for any queries related to the assignment and avoid asking queries on the last day.
3. There will be a live demonstration and viva to evaluate the fourth assignment.

1 Goals

Implement user-level threads in Linux from scratch. It should be a user-level threading library that aims to support the major functions of the pthread library. The kernel's involvement should be minimal. The following facilities must be provided:

- Per-thread state
- Thread-level stacks
- Support for thread suspension, scheduling and context switches
- Different scheduling priorities (at least two different schedulers)
- Mutexes, semaphores and reader-writer locks

2 Submission Instructions

- We will run MOSS on the submissions. Any cheating will result in a zero in the assignment, a penalty as per the course policy, and possibly much stricter penalties (including a fail grade).
- Prepare a PDF file named **A4_report.pdf** that includes a description of the threading library and the design choices for your implementation.

How to submit:

1. Copy your report to the xv6 root directory.

2. Create a zip file that contains the report, test scripts, and a folder that contains the files for user-level threading, and then name the zip file as $assignment4\_hard\_ < entryNumber > .zip$. Submit this zip file to Moodle. Entry number format: 2020CSZ2445. *Note that all English letters are in capitals.*